Xiao-Yang Liu, Ziyi Xia, Jingyang Rui, Jiechao Gao, Hongyang Yang, Ming Zhu, Christina Wang, Zhaoran Wang, and Jian Guo. 2022. Finrl-meta: Market environments and benchmarks for data-driven financial reinforcement learning. *Advances in Neural Information Processing Systems*, 35:1835–1849.

Arman Khadjeh Nassirtoussi, Saeed Aghabozorgi, Teh Ying Wah, and David Chek Ling Ngo. 2014. Text mining for market prediction: A systematic review. *Expert Systems with Applications*, 41(16):7653–7670.

Raj Sanjay Shah, Kunal Chawla, Dheeraj Eidnani, Agam Shah, Wendi Du, Sudheer Chava, Natraj Raman, Charese Smiley, Jiaao Chen, and Diyi Yang. 2022. When flue meets flang: Benchmarks and large pre-trained language model for financial domain. *arXiv preprint arXiv:2211.00083*.

Si Shi, Jianjun Li, Guohui Li, and Peng Pan. 2019. A multi-scale temporal feature aggregation convolutional neural network for portfolio management. In *Proceedings of the 28th ACM international conference on information and knowledge management*, pages 1613–1622.

Shuo Sun, Rundong Wang, and Bo An. 2023. Reinforcement learning for quantitative trading. *ACM Transactions on Intelligent Systems and Technology*, 14(3):1–29.

Andrew Szot, Max Schwarzer, Harsh Agrawal, Bogdan Mazoure, Rin Metcalf, Walter Talbott, Natalie Mackraz, R Devon Hjelm, and Alexander T Toshev. 2023. Large language models as generalizable policies for embodied tasks. In *The Twelfth International Conference on Learning Representations*.

Kieran Wood, Sven Giegerich, Stephen Roberts, and Stefan Zohren. 2021. Trading with the momentum transformer: An intelligent and interpretable architecture. *arXiv preprint arXiv:2112.08534*.

Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhanjan Kambadur, David Rosenberg, and Gideon Mann. 2023. Bloomberggpt: A large language model for finance. *arXiv preprint arXiv:2303.17564*.

Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, Yueru He, Mengxi Xiao, Dong Li, Yongfu Dai, Duanyu Feng, and 1 others. 2024a. The finben: An holistic financial benchmark for large language models. *arXiv preprint arXiv:2402.12659*.

Qianqian Xie, Weiguang Han, Xiao Zhang, Yanzhao Lai, Min Peng, Alejandro Lopez-Lira, and Jimin Huang. 2023. Pixiu: A large language model, instruction data and evaluation benchmark for finance. *arXiv preprint arXiv:2306.05443*.

Qianqian Xie, Dong Li, Mengxi Xiao, Zihao Jiang, Ruoyu Xiang, Xiao Zhang, Zhengyu Chen, Yueru He, Weiguang Han, Yuzhe Yang, and 1 others. 2024b. Open-finllms: Open multimodal large language models for financial applications. *arXiv preprint arXiv:2408.11878*.

Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023a. Fingpt: Open-source financial large language models. *arXiv preprint arXiv:2306.06031*.

Hongyang Yang, Boyu Zhang, Neng Wang, Cheng Guo, Xiaoli Zhang, Likun Lin, Junlin Wang, Tianyu Zhou, Mao Guan, Runjia Zhang, and 1 others. 2024. Finrobot: An open-source ai agent platform for financial applications using large language models. *arXiv preprint arXiv:2405.14767*.

Yi Yang, Yixuan Tang, and Kar Yan Tam. 2023b. Investlm: A large language model for investment using financial domain instruction tuning. *arXiv preprint arXiv:2309.13064*.

Yi Yang, Mark Christopher Siy Uy, and Allen Huang. 2020. Finbert: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*.

Yunan Ye, Hengzhi Pei, Boxin Wang, Pin-Yu Chen, Yada Zhu, Ju Xiao, and Bo Li. 2020. Reinforcement-learning based portfolio management with augmented asset movement prediction states. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 1112–1119.

Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W Suchow, and Khaldoun Khashanah. 2024a. Finmem: A performance-enhanced llm trading agent with layered memory and character design. In *Proceedings of the AAAI Symposium Series*, volume 3, pages 595–597.

Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyang Deng, Yupeng Cao, Zhi Chen, Jordan W Suchow, Rong Liu, Zhenyu Cui, Zhaozhuo Xu, and 1 others. 2024b. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. *arXiv preprint arXiv:2407.06567*.

Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyang Deng, Yuechen Jiang, Yupeng Cao, Zhi Chen, Jordan Suchow, Zhenyu Cui, Rong Liu, and 1 others. 2025. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. *Advances in Neural Information Processing Systems*, 37:137010–137045.

Yuexiang Zhai, Hao Bai, Zipeng Lin, Jiayi Pan, Shengbang Tong, Yifei Zhou, Alane Suhr, Saining Xie, Yann LeCun, Yi Ma, and 1 others. 2024. Fine-tuning large vision-language models as decision-making agents via reinforcement learning. *arXiv preprint arXiv:2405.10292*.

Wentao Zhang, Lingxuan Zhao, Haochong Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, and 1 others. 2024. Finagent: A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist. *arXiv preprint arXiv:2402.18485*.

Yuzhe Zhang and Hong Zhang. 2023. Finbert–mrc: financial named entity recognition using bert under the machine reading comprehension paradigm. *Neural Processing Letters*, 55(6):7393–7413.

Zihao Zhang, Stefan Zohren, and Stephen J. Roberts. 2019. Deep reinforcement learning for trading. In *The Journal of Financial Data Science*.

A Additional Algorithmic Details: FLAG-TRADER with PPO

In this section, we outline a detailed procedure for training the FLAG-TRADER architecture via PPO, where the POLICY_NET (actor) and the VALUE_NET (critic) share a subset of trainable parameters from a LLM, with $\theta = \big(\theta_{\texttt{train}}, \theta_P, \theta_V\big)$. We define $\theta_{policy} = \big(\theta_{\texttt{train}}, \theta_P\big)$ and $\theta_{value} = \big(\theta_{\texttt{train}}, \theta_V\big)$ for simplicity.

Advantage Estimation. We use the Generalized Advantage Estimation (GAE) to compute the advantage function A_t:

$$A_t \;=\; \sum_{k=0}^{T-1} (\gamma\lambda)^k \big[r_{t+k} + \gamma V_{\theta_{value}}(s_{t+k+1}) - V_{\theta_{value}}(s_{t+k})\big], \tag{17}$$

where γ is the discount factor, and λ is the GAE parameter.

Probability Ratio. Let $\theta_{policy,\text{old}}$ denote the parameters before the current update. The PPO probability ratio is

$$r_t(\theta_{policy}) \;=\; \frac{\pi_{\theta_{policy}}(a_t \mid s_t)}{\pi_{\theta_{policy,\text{old}}}(a_t \mid s_t)}. \tag{18}$$

PPO Clipped Objective. PPO clips this ratio to prevent overly large updates. The surrogate objective is

$$\mathcal{L}_P(\theta_{policy}) \;=\; \mathbb{E}_t\Big[\min\big(r_t(\theta_{policy})\, A_t,\; \text{clip}\big(r_t(\theta_{policy}),\, 1-\varepsilon,\, 1+\varepsilon\big)\, A_t\big)\Big], \tag{19}$$

where ε is a hyperparameter.

Value Function Loss. The critic (value network) is updated by minimizing the difference between the predicted value $V_{\theta_{value}}(s_t)$ and the target return R_t. A common choice is:

$$\mathcal{L}_V(\theta_{value}) \;=\; \mathbb{E}_t\Big[(V_{\theta_{value}}(s_t) - R_t)^2\Big]. \tag{20}$$

Combined Loss. We often add an entropy term to encourage exploration, yielding the overall objective:

$$\mathcal{L}_{\text{total}}(\theta) \;=\; -\,\mathcal{L}_P(\theta_{policy}) \;+\; c_1\, \mathcal{L}_V(\theta_{value}) \;-\; c_2\, \mathcal{H}\big(\pi_{\theta_{policy}}\big), \tag{21}$$

where c_1 and c_2 are weighting coefficients, and $\mathcal{H}(\pi_{\theta_{policy}})$ represents the policy entropy.

Parameter Updates. At each iteration, we apply gradient descent on the total loss:

$$\theta_P \leftarrow \theta_P \;-\; \eta\, \nabla_{\theta_P} \mathcal{L}_P, \tag{22}$$

$$\theta_V \leftarrow \theta_V \;-\; \eta\, \nabla_{\theta_V} \mathcal{L}_V, \tag{23}$$

$$\theta_{\texttt{train}} \leftarrow \theta_{\texttt{train}} - \beta\, \nabla_{\theta_{\text{train}}} \mathcal{L}_{\text{total}}, \tag{24}$$

where η and β are learning rates for the policy head, value head, and trainable LLM layers respectively. The algorithm is summerized in Algorithm 2.